



Práctica 3: Sistema Cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Table of Contents

Información general	1
Datos de la simulación	1
Hipotenso	2
Hipertenso	3
Funcion	3
Funcion	4

Información general



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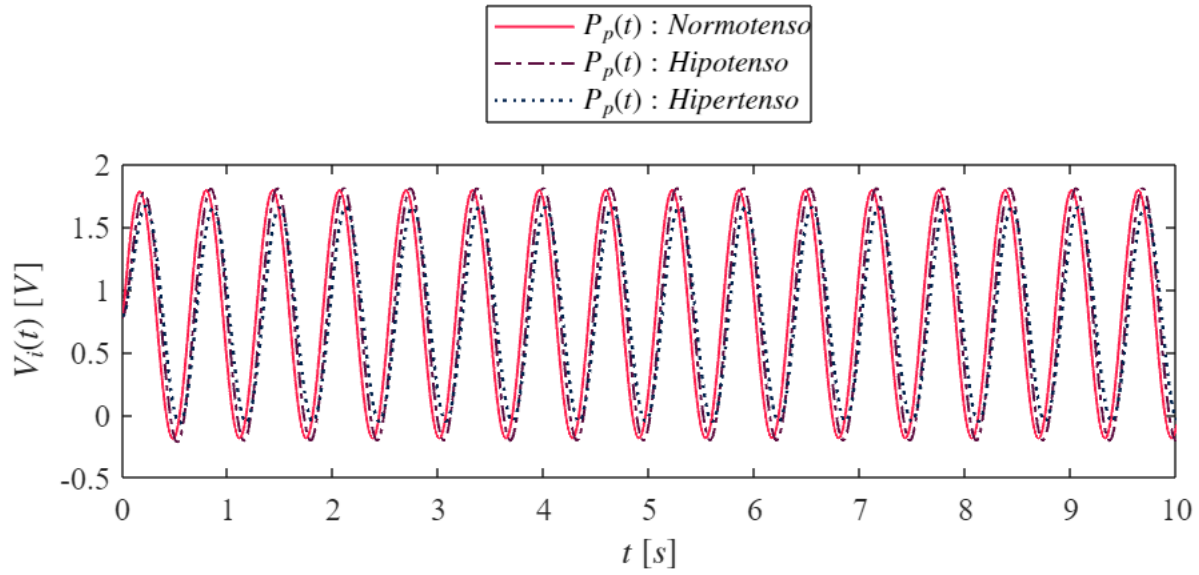
Modelado de Sistemas Fisiológicos

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Datos de la simulación

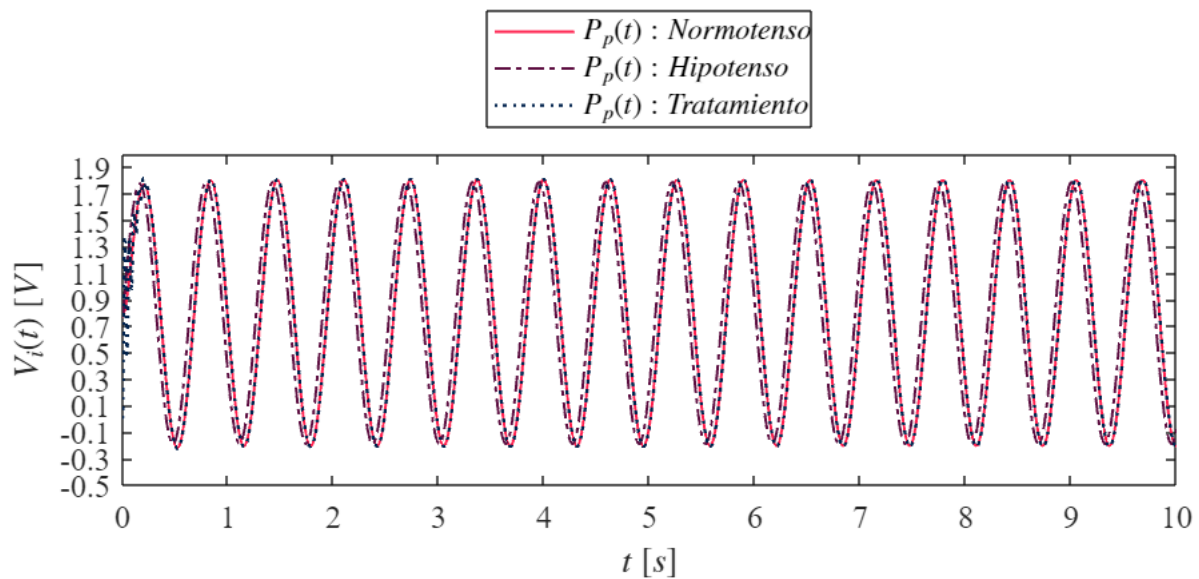
```
clc; clear; close all; warning('off','all')
tend = '10';
file = 'Sistemacar';
open_system(file);
parameters.StopTime = tend;
parameters.Solver = 'ode15s';
parameters.MaxStep = '1E-3';
```

```
x = sim (file, parameters);
plotsignal1 (x.t, x.Ppx, x. Ppy, x.Ppz)
```



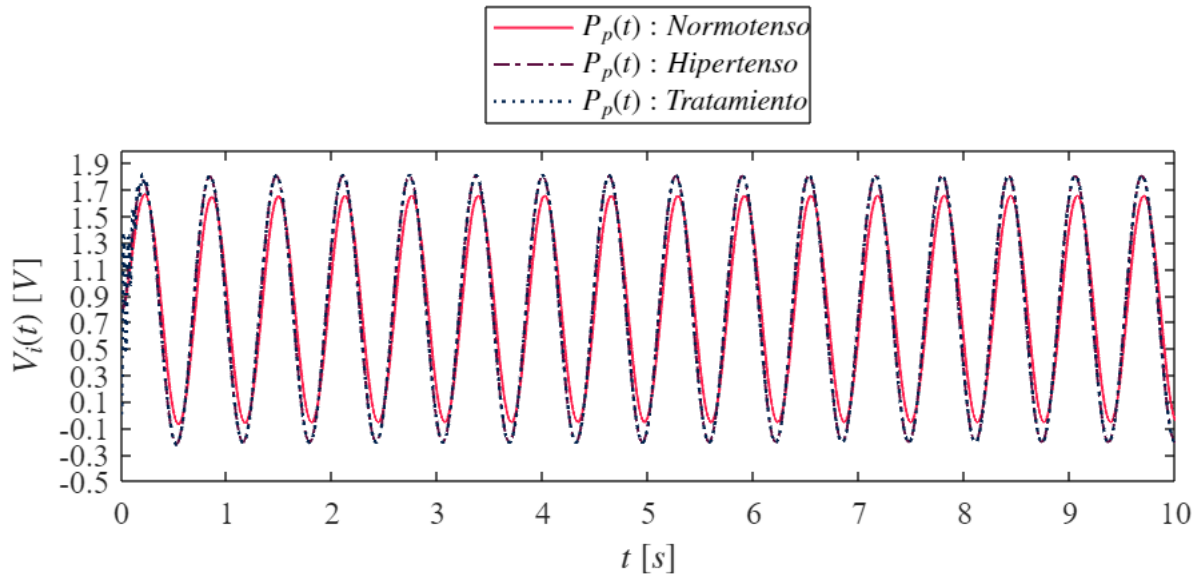
Hipotenso

```
file = 'Sistemcar2';
open_system(file);
x = sim (file, parameters);
opt = 'Hipotenso';
plotsignal (x.t, x.Ppx, x. Ppy, x.Ppz, opt)
```



Hipertenso

```
file = 'Sistemacar2hipe';  
open_system(file);  
x = sim (file, parameters);  
opt = 'Hipertenso';  
plotsignal (x.t, x.Ppx, x.Ppy, x.Ppz, opt)
```



Funcion.

```
function plotsignal1(t, Ppx, Ppy, Ppz)  
set(figure(), 'Color', 'w')  
set(gcf, 'Units', 'Centimeters', 'Position',[1,1,18,8])  
set(gca, 'FontName', 'Times New Roman')  
fontsize(12, 'points')  
rosa = [255/255, 32/255, 78/255];  
morado = [93/255, 14/255, 65/255];  
azul = [0/255, 34/255, 77/255];  
hold on; grid off; box on  
  
t = t(:);  
% Convertir objetos timeseries a arreglos numéricos  
if isa(Ppx, 'timeseries'), Ppx = Ppx.Data(:); end  
if isa(Ppy, 'timeseries'), Ppy = Ppy.Data(:); end  
if isa(Ppz, 'timeseries'), Ppz = Ppz.Data(:); end  
  
plot(t, Ppx, 'LineWidth', 1, 'Color', rosa)
```

```

plot(t, Ppy, '-.', 'LineWidth', 1, 'Color', morado)
plot(t, Ppz, ':', 'LineWidth', 1.2, 'Color', azul)
xlabel('$t$ $[s]$', 'Interpreter', 'Latex')
ylabel('$V_i(t)$ $[V]$', 'Interpreter', 'Latex')
L = legend('$P_p(t)$: Normotenso$', '$P_p(t)$: Hipotenso$', '$P_p(t)$:Hipertenso$');
set(L, 'Interpreter', 'latex', 'Location', 'northoutside', 'Box', 'on')
end

```

Funcion

```

function plotsignal(t, Ppx, Ppy, Ppz, opt)
set(gcf, 'Color', 'w')
set(gcf, 'Units', 'Centimeters', 'Position', [1,1,18,8])
set(gca, 'FontName', 'Times New Roman')
fontSize(12, 'points')
rosa = [255/255, 32/255, 78/255];
morado = [93/255, 14/255, 65/255];
azul = [0/255, 34/255, 77/255];
hold on; grid off; box on

t = t(:);
% Convertir objetos timeseries a arreglos numéricos
if isa(Ppx, 'timeseries'), Ppx = Ppx.Data(:); end
if isa(Ppy, 'timeseries'), Ppy = Ppy.Data(:); end
if isa(Ppz, 'timeseries'), Ppz = Ppz.Data(:); end

plot(t, Ppx, 'LineWidth', 1, 'Color', rosa)
plot(t, Ppy, '-.', 'LineWidth', 1, 'Color', morado)
plot(t, Ppz, ':', 'LineWidth', 1.2, 'Color', azul)
xlabel('$t$ $[s]$', 'Interpreter', 'Latex')
ylabel('$V_i(t)$ $[V]$', 'Interpreter', 'Latex')
if opt == "Hipotenso"
L = legend('$P_p(t)$: Normotenso$', '$P_p(t)$: Hipotenso$', '$P_p(t)$:Tratamiento$');
set(L, 'Interpreter', 'latex', 'Location', 'northoutside', 'Box', 'on')

xlim([0, 10]); xticks(0:1:10)
ylim([-0.5, 2]); yticks(-0.5:0.2:2)
elseif opt == "Hipertenso"
L = legend('$P_p(t)$: Normotenso$', '$P_p(t)$: Hipertenso$', '$P_p(t)$:Tratamiento$');
set(L, 'Interpreter', 'latex', 'Location', 'northoutside', 'Box', 'on')
xlim([0, 10]); xticks(0:1:10)
ylim([-0.5, 2]); yticks(-0.5:0.2:2)
end

exportgraphics(gcf, [opt, '.pdf'], 'ContentType', 'Vector')
exportgraphics(gcf, [opt, '.png'], 'Resolution', 600);
print(opt, '-dsvg');
print(opt, '-depsc');
end

```

